

HM Ziyad

Software Engineer and AI Specialist

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PROFILE

I am a Software Engineer and AI Specialist dedicated to building intelligent, scalable systems. My expertise ranges from low-level systems programming in C++ to developing modern, full-stack web applications using Next.js and TypeScript. My current focus is on Generative AI and RAG Pipelines, leveraging Large Language Models to solve complex real-world problems (Medhabi.AI). I maintain a strong interest in competitive programming and algorithmic optimization.

EDUCATION

Bachelor of Science in CSE

Ahsanullah University of Science and Technology
2025

CGPA: 3.44

Higher Secondary School Certificate

Saint Joseph Higher Secondary School
2020

GPA: 5.00

SKILLS

Programming Language

C/C++, Java, Python, C#/.NET Core, PHP, dart

AI/ML

PyTorch, Tensorflow, Embedding Algorithms

Database

SQL(MySQL, MSSQL), NoSQL(Firebase, Mongo DB)

IoT

Arduino, ESP, Raspberry-Pi, Sensors

EXPERIENCE

05/12/2025 –
01/08/2026

AI Engineer

Sparktech IT Limited

Improved Generative AI and RAG Pipelines for AI applications, resulting in a 17% reduction in query response time.

04/11/2025 –
05/12/2025

Intern

City Bank PLC, Enterprise Architecture Division (IT)

Collaborated on enterprise SDLC and UX enhancements to streamline workflows for 150+ banking employees.

01/11/2024 –
15/11/2025

Undergraduate Research Assistant

AUST Satellite Lab

Engineered a robust radio communication system for the AUSTSAT (Nano Satellite), achieving a 57% increase in image transfer speed using RF modules and specialized encoding/decoding techniques.

ACHIEVEMENTS

Achieved a top 15 ranking out of 550 Teams in Solvio AI Hackathon 2025 hosted by Sheba Platform
Sheba Platform, Programming Hero

Achieved a top 10 ranking out of 100 teams in the SUST inter-university hackathon 2023.
SUST CSE Society

Leadership Award

AUST Association for students welfare

1st runners up at Intra AUST programming contest junior category fall-2021
AUST CSE Society

2nd runners up at Intra AUST programming contest senior category spring-2022
AUST Programming and Informatics Club

2nd Position ACM lab selection contest
AUST CSE Society

POSITION OF RESPONSIBILITY

01/06/2025 –
01/15/2026

AUST Programming and Informatics Club

President

Organized IUPC(Inter University Programming Contest) & Educated 300+ students on AI/ML and CNCF, resulting in a 70% increase in club engagement. Organized She-Stem National Job Fair.

11/07/2025 –
01/10/2026

AUST Satellite Lab

Head | Embedded System and Programming

Designed Satellite Circuits, Embedded programming, Onboard Sensors Integration and Developed an Image encoding and decoding technique* for Long range radio transmission.

01/02/2025 –
13/04/2025

AUST CSE Society

Organizer

Legacy event of CSE department, AUST. Organized IAPC-Intra AUST Programming Contest, Code Refactoring Contest, Software Exhibition and many more segments.

2019 – 2020

Josephite Math Club

Executive Committee

Organized a nation-wide Math Olympiad in Saint Joseph Higher Secondary School.

2018 – 2019

Scintilla Science Club

Executive Committee

PROJECTS

LeapSync - Humanoid Model [🔗](#)

Three.js, Eleven Labs

This is a cutting-edge web application that features a 3D avatar capable of real-time lip-syncing. Built with modern web technologies, it allows users to type text and watch the avatar speak it back with natural facial animations and punctuation-aware pausing.

Medhabi.AI [🔗](#)

RAG Pipeline

This project implements a Retrieval-Augmented Generation (RAG) system to answer physics questions based on the NCTB Physics 1st Paper (Bengali) book.

Calm Mind [🔗](#)

LLM, AI-Powered

Calm Mind is an intelligent, AI-powered meditation application designed to generate personalized meditation sessions based on your current mood and needs. Built with Next.js 14, Tailwind/Custom CSS, and Google Gemini AI, it offers a seamless, beautiful experience to help you find peace in a busy world.

Open IUMS - Open Credit University Management System [🔗](#)

SDLC, Software Architect

Architected a scalable, efficient university management software for open credit systems, applying SDLC principles for requirements analysis, database design, and modular architecture. Focused on usability, maintainability, and scalability. Flutter

AUST Overflow - Q&A Platform for University Students [🔗](#)

Software Design

Designed a real-time, responsive question-and-answer platform using ASP.NET MVC, JavaScript, and Firebase. Managed full-stack development, focusing on secure file handling, modular components (authentication, Q&A, moderation), and robust data management.

Silent Signal - Embedded System for Noiseless Horn System [🔗](#)

IoT

A low-cost, scalable Arduino-based embedded system using NRF modules to detect car horn noise pollution and alert nearby vehicles, promoting responsible honking for smart city traffic.

LightChain - A blockchain based IoT security protocol [🔗](#)

IoT

LightChain, an IoT blockchain using ESP32 nodes and a Raspberry Pi Gateway for enhanced security. Utilized SHA-256 and OpenSSL for immutable, auditable, and efficiently retrievable data. Integrated Post-Quantum Cryptography (PQC) for future-proof security with minimal smart device overhead.